

Lab Activity Report: Introduction to AWS IAM – AWS Cloud Security (50 points)
ITELEC 4103

Overview:

This lab focuses on the practical use of **AWS Identity and Access Management (IAM)** to manage user permissions and control access to AWS resources. By exploring key IAM functionalities such as user and group creation, policy inspection, and service access management, participants will understand how IAM enhances security in cloud environments.

REPORT TEMPLATE:

The report **should not exceed 6 pages**. For each additional page beyond the limit, a 5-point deduction will be applied.

Please use screenshots to enhance your explanations or clarify specific points. Screenshots should be used to support the discussion and illustrate key concepts.

I. Brief Rationale for the Activity Conducted:

Provide a short introduction explaining the purpose of this activity. Discuss why learning IAM is essential for cloud security and how it aids in effectively managing AWS accounts, permissions, and resource access.

II. Discussion (based on tasks performed):

- **Creating IAM Users and Groups:**
 - Explain the steps taken to create IAM users and groups. Discuss why organizing users into groups helps streamline permissions.
 - **Inspecting Policies:**
 - Describe the process of inspecting AWS policies (e.g., managed and custom policies). Highlight the key components of a policy document, such as actions, resources, and conditions.
 - **Adding Users/Groups:**
 - Provide details on adding users to groups and assigning appropriate permissions. Explain how this step ensures role-based access control.
 - **Using the IAM URL:**
 - Discuss the significance of IAM user URLs and how they are used to grant access to the AWS Management Console.
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III. Exploring Effects of Policies on Service Access:

Present and discuss the effects of applying policies to your IAM users or groups.

- **Example 1:** Show how limiting access to a specific AWS service (e.g., S3 or EC2) affects users' abilities to interact with those services.
- **Example 2:** Demonstrate how a "Deny" policy prevents users from performing certain actions, despite having broad access permissions.
- **Example 3:** Show how attaching a "Read-Only" policy affects users' ability to view but not modify resources.
- **Etc.**

IV. Key Learnings:

List and elaborate on your **three (3) significant learnings** from the activity.

RUBRICS FOR THE ACTIVITY

	3 pts	6 pts	10 pts	12.5 pts
Focus	The discussion lacked clarity and direction, with little to no connection to the main topic or activity.	Some topics were addressed, but the discussion was not comprehensive or lacked depth in several areas.	All relevant topics were addressed with a clear and organized focus, though some areas may have lacked detail.	Each topic was addressed with thorough, detailed, and well-organized discussion, providing a deep focus on the key areas.
Clarity	The answer or discussion was unclear and difficult to follow, lacking a direct connection to the topic.	The answer or discussion addressed the topic, but in a roundabout or vague manner, making it harder to understand.	The answer or discussion directly addressed the topic, providing clarity and coherence.	The answer or discussion was exceptionally clear, concise, and to the point, making the topic easy to understand.
Organization	The answer or discussion had no clear structure, leaving gaps for readers to fill in.	The answer or discussion had some structure and painted a complete picture, but the organization was not orderly, making it less effective.	Details were presented in a logical order, but the structure could have been more engaging. Some loose ends remained, and the flow of the discussion was somewhat uneven.	Details were logically ordered and presented in engaging ways. The discussion flowed smoothly, covering all main points clearly, and concluded effectively.
Application	The answer or discussion showed a lack of understanding of basic concepts and principles, with no clear connection to the material.	The answer or discussion was able to state a few concepts and principles but did not fully apply them.	The answer showed a clear understanding and application of basic concepts and principles, demonstrating comprehension of the material.	The answer integrated and applied concepts and principles from the module materials effectively, showing a deep understanding and practical application of the ideas.